

1. One of the solutions for x in the equation $x^2 + bx + c = 0$ where b and c are integers is $x = 1 + \sqrt{2}$. Find the value of the sum $(b + c)$.
2. $i = \sqrt{-1}$. Solve for b if $1 - \frac{b}{4 + i\sqrt{3}} + \frac{b}{4 - i\sqrt{3}} = 0$.
3. The graph of $4x^2 - y^2 - 8x + 2y = 1$ is rotated 90° clockwise about the origin. Find the coordinates of the center of the new conic. Express your answer as the ordered pair (x, y) .
4. Find **the number** of polar points below that are coincident (occupy the same position) as the polar point $\left(5, \frac{\pi}{6}\right)$?
I. $\left(-5, \frac{7\pi}{6}\right)$ II. $\left(5, \frac{5\pi}{6}\right)$ III. $\left(5, \frac{13\pi}{6}\right)$ IV. $\left(5, \frac{11\pi}{6}\right)$ V. $\left(-5, \frac{-\pi}{6}\right)$
5. Find the value(s) of x for which $4 + 3\sqrt{7 - x} = x + 7$.
6. Simplify the expression $(\cos^6 x + 3(\cos^4 x)(\sin^2 x) + 3(\cos^2 x)(\sin^4 x) + \sin^6 x)$.
7. Let $P(x) = 0$ be a third degree polynomial equation with **integer** coefficients and leading coefficient of 1. Let the polynomial equation $P(x) = 0$ have at least one root that is an **integer**. If $P(6) = -11$, $P(4) = 3$, and $P(-4) = -341$, find $P(10)$.